

The addition rule

The general addition rule subtracts the overlap counted in both events.

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

If $P(A) = 0.4$, $P(B) = 0.5$, and $P(A \text{ AND } B) = 0.2$, what is $P(A \text{ OR } B)$?

Why subtract the overlap?

Synthetic survey: 40 laptop users, 25 tablet users, and 10 use both.

What is wrong with adding $40 + 25$ to count those using either device?

A union from counts

Synthetic survey of 100 students: 40 use laptops, 25 use tablets, and 10 use both. Select one uniformly.

What is the probability of laptop or tablet use?

A disjoint union

A fair six-sided die is rolled.

What is the probability of rolling 1 or 6?

Complement of a union

$$P(A \text{ OR } B) = 0.65.$$

What is the probability of neither A nor B?

An intersection from a union

$P(A) = 0.55$, $P(B) = 0.40$, and $P(A \text{ OR } B) = 0.75$.

What is $P(A \text{ AND } B)$?

The multiplication rule

The general multiplication rule uses a conditional probability.

$$P(A \cap B) = P(A)P(B | A)$$

If $P(A) = 0.4$ and $P(B | A) = 0.3$, what is $P(A \text{ AND } B)$?

A dependent product

A box has 3 red and 2 blue tokens. Draw two uniformly without replacement.

What is the probability both are red?

A different order

A box has 3 red and 2 blue tokens. Draw two uniformly without replacement.

What is the probability the first is red and the second blue?

Either order

A box has 3 red and 2 blue tokens. Draw two uniformly without replacement.

What is the probability of one red and one blue in either order?

An independent product

Two independent fair dice are rolled.

What is the probability both show 6?

At least one

Two independent fair dice are rolled.

What is the probability at least one shows 6?

Exactly one

Two independent fair dice are rolled.

What is the probability exactly one shows 6?

A conditional ratio

Conditional probability uses the intersection divided by the probability of the conditioning event.

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

If $P(A \text{ AND } B) = 0.12$ and $P(B) = 0.30$, what is $P(A \mid B)$?

A zero denominator

$$P(B) = 0.$$

Can the usual conditional-probability ratio define $P(A \mid B)$?

Read the notation

A = laptop use; B = tablet use.

How would you describe $P(A \text{ AND } B)$ in words?

Choose the rule

A probability question asks whether at least one of two events occurs.

Which rule naturally connects the event probabilities and their overlap?

Check coherence

A proposed model has $P(A) = 0.2$ and $P(A \text{ AND } B) = 0.3$.

Can these probabilities describe valid events?

Three independent events

Three independent transmissions each succeed with probability 0.9.

What is the probability all three succeed?

A complement shortcut

Three independent transmissions each succeed with probability 0.9.

What is the probability at least one fails?